CAMPIFY

Micro Project Report

Submitted for the Partial Fulfillment of the Requirements

for the Award of the Degree of

Master of Computer Applications

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BONAFIDE CERTIFICATE

This is to certify that the Micro Project Report entitled “CAMPIFY” has been submitted by the students listed below to Muthoot Institute of Technology and Science, Varikoli, for the partial fulfillment of the requirements for the award of the degree of Master of Computer Applications (MCA), and is a bonafide record of the project work carried out by them under our supervision and guidance during the academic year 2026–28.

The Micro Project was carried out as part of the First Semester MCA course WEB PROGRAMMING LAB.

Place: Varikoli

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Submitted for the Final Evaluation held on ……

Name and Signature of the Examiner

DECLARATION

We, the undersigned, hereby declare that the Micro Project Report “Campify”, submitted for the partial fulfilment of the requirements for the award of degree of Master of Computer Applications of the APJ Abdul Kalam Technological University, Kerala is a bonafide work done by us under the supervision of Dr Sujitra Shankar. This submission represents our ideas in our own words. We also declare that we have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in our submission. This report has not been previously formed the basis for the award of any degree, diploma or similar title of any other University.

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Place: Varikoli

Date:

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ABSTRACT

Campify is a web-based campus marketplace designed to help students buy, sell, and exchange student-to-student services within their college community. Students often depend on informal messaging groups or notice boards to find used books, electronic devices, room accessories, and academic help. These approaches make it difficult to search, verify, save, and manage listings in one place.

The system provides a focused marketplace for authenticated college students. Users register with an institutional email address, sign in securely, browse active listings, search and filter by category, save favorite listings, enquire through the seller's campus email, and create their own item or student-help posts. A profile area allows users to review their posts, delist listings, change passwords, and delete their account.

The frontend is implemented using HTML, CSS, and JavaScript. PHP provides the server-side API and session handling, while Supabase Authentication and PostgreSQL provide identity and persistent data storage. Row-level security policies restrict access to authenticated students and ensure that users can modify only their own listings and favorites. Input validation, email-domain restrictions, session cookies, and server-side checks improve reliability and security.

Campify demonstrates the practical use of client-server architecture, authentication, CRUD operations, REST-style API communication, relational database design, and responsive user-interface development. The current version focuses on the core campus marketplace experience; payment processing, delivery, chat, moderation, and mobile applications are outside the scope of this project.

1. INTRODUCTION

1.1 Introduction and Background

A college campus contains a small but active local economy. Students frequently need textbooks, calculators, furniture, electronics, clothing, tutoring, design support, and project guidance. At the same time, many students have useful items or skills that they would like to share with others. General online marketplaces are often too broad, while informal social-media groups do not provide structured search, ownership, or account management.

Campify addresses this gap by providing a campus-only marketplace. The system combines a simple browsing interface with authenticated posting and a persistent backend. Its design emphasizes local discovery, student identity, and safe communication rather than anonymous public commerce.

1.2 Problem Statement

Students do not have a dedicated, searchable, and structured platform for exchanging used items and student help within their campus. Existing informal methods make it difficult to find relevant posts, distinguish active listings, contact the correct student, save interesting items, and remove or manage old posts. A secure web application is therefore required to centralize campus listings while restricting access to verified institutional users.

1.3 Objectives of the Project

• To develop a user-friendly campus marketplace for students.

• To restrict registration and access to valid college email addresses.

• To allow users to create, view, search, filter, save, and enquire about listings.

• To support both items for sale and student-help offers or requests.

• To provide profile-based listing management and account settings.

• To store marketplace data persistently using Supabase PostgreSQL.

• To apply server-side validation and row-level security for reliable access control.

1.4 Scope of the Project

The scope includes registration, email confirmation, sign-in, session management, marketplace browsing, category filtering, text search, listing creation, image upload as a data URL, favorites, enquiry through Gmail, profile listing management, delisting, password change, and account deletion. The scope does not include online payments, delivery tracking, real-time chat, administrator moderation, recommendation algorithms, or a native mobile application.

2. SYSTEM ANALYSIS

2.1 Existing System

The existing approach generally uses classroom groups, messaging applications, notice boards, or general-purpose marketplace sites. These methods are useful for quick communication but do not provide a consistent listing format or a single searchable catalogue. Posts become difficult to locate as new messages arrive, and old or unavailable posts may remain visible.

Existing approach	Limitations
Messaging groups	Posts are mixed with unrelated conversations and are difficult to search.
Physical notice boards	Limited reach, no live updates, and no structured contact workflow.
General marketplaces	Not focused on campus membership or student-help exchanges.
Informal direct sharing	No persistent listing management or favorites.

2.2 Proposed System

Campify provides a dedicated, authenticated campus marketplace. Users see only active listings after signing in. The interface separates items for sale from student help, supports category chips and keyword search, and presents seller information and enquiry actions. The backend validates all important fields and stores records in Supabase with row-level security.

Feature	Benefit
Institutional email authentication	Restricts the service to the intended college community.
Structured listing form	Ensures that title, description, category, price, and type are available.
Search and filtering	Helps students locate relevant posts quickly.
Favorites	Lets users keep track of listings for later reference.
Profile management	Allows users to delist posts and manage account settings.
Database security policies	Prevents users from changing another user’s records.

2.3 Feasibility Study

Technical feasibility: The project uses widely supported web technologies and a hosted backend with authentication and PostgreSQL storage. The application can run locally using PHP 8.0 or later with cURL and mbstring extensions.

Operational feasibility: The interface follows familiar marketplace patterns and requires no special training. Students can browse listings, post an item, and manage their account from a normal web browser.

Economic feasibility: The implementation uses open-source frontend and server technologies. Supabase provides the required authentication and database services for a student-scale prototype, subject to its service limits and configuration.

3. SYSTEM DESIGN AND IMPLEMENTATION

3.1 System Requirements

3.1.1 Hardware Requirements

Component	Minimum requirement
Processor	Dual-core processor or better
Memory	4 GB RAM
Storage	At least 500 MB free space for source files and tools
Display	1366 × 768 resolution or better
Network	Internet connection for Supabase authentication and data access

3.1.2 Software Requirements

Software / technology	Purpose
HTML5 and CSS3	Structure and responsive presentation
JavaScript	Client-side interaction, validation, and API communication
PHP 8.0+	Server-side API and session handling
Supabase Auth	Email/password authentication and identity management
Supabase PostgreSQL	Persistent listings and favorites storage
Modern web browser	Application execution and user interaction
Visual Studio Code	Development and project maintenance

3.2 System Architecture

Campify follows a three-layer web architecture. The presentation layer consists of HTML pages, CSS styles, and JavaScript modules. The application layer consists of PHP API endpoints that validate requests, manage sessions, and communicate with Supabase. The data layer consists of Supabase Authentication and PostgreSQL tables protected by row-level security.

Layer	Main components	Responsibility
Presentation	index.html, login.html, profile.html, styles.css, app.js	Render pages, collect input, display results, and provide interaction.
Application	api/auth.php, listings.php, favorites.php, profile.php, delist.php	Validate requests, authorize users, manage sessions, and return JSON responses.
Data	Supabase Auth, listings, listing_favorites	Store user identities, marketplace records, and saved listings.

Typical request flow: (1) the browser sends a request to a PHP endpoint; (2) the endpoint reads the HTTP-only session and access token; (3) the endpoint validates the payload; (4) PHP sends an authenticated request to Supabase; (5) Supabase applies database policies; and (6) the endpoint returns a success or error response for the interface to display.

3.3 Functional Modules and Workflow

3.3.1 Authentication Module

The authentication module supports registration, sign-in, sign-out, email confirmation, and session lookup. Registration accepts only valid @mgits.ac.in addresses and stores the access and refresh tokens in a server-side PHP session. Confirmation links are handled by auth-confirmation.html and auth-confirmation.js. Password validation and clear error responses are applied before requests are sent to Supabase.

3.3.2 Marketplace Module

The marketplace module displays active listings and separates them into “For sale” and “Student help” views. Users can search by title, description, category, or seller name, and can filter by books, technology, room and living, style, or help. Listing cards show the title, price or reward, description, seller, time, tag, image or placeholder art, and an Enquire action.

3.3.3 Listing and Favorites Module

Authenticated users can create an item listing or a student-help offer/request. The server validates the type, direction, title length, description length, category, price, and optional image format and size. Favorites are stored separately and are restricted to the signed-in user by row-level security.

3.3.4 Profile and Account Module

The profile page displays the signed-in user's listings and account settings. A user can delist a post so that it no longer appears in the active marketplace while retaining the record with a timestamp. The account settings allow password changes after verification of the current password and account deletion after explicit confirmation.

3.4 Database Design

The primary marketplace table is listings. It stores the owner, listing type, direction, title, description, price, category, seller information, optional image data, creation time, and delisting time. The listing_favorites table maps a user to a listing and uses a composite primary key to prevent duplicate favorites.

Table / object	Important fields	Purpose
auth.users	id, email, password identity	Managed by Supabase Auth for user accounts.
listings	id, owner_id, kind, title, description, price, category, created_at, delisted_at	Stores items and student-help listings.
listing_favorites	owner_id, listing_id, created_at	Stores each user's saved listings.
enforce_campus_email()	email validation trigger	Rejects accounts outside the college email domain.
delete_own_account()	authenticated user context	Deletes only the currently authenticated user's records.

3.5 Implementation Details

3.5.1 Registration and Login

The login interface switches between sign-in and create-account modes. The JavaScript client sends JSON to api/auth.php. The endpoint validates the email and password, calls Supabase Auth, and stores the returned session in an HTTP-only server session. Users who require confirmation are directed to the email confirmation page.

3.5.2 Creating and Reading Listings

The post dialog collects the kind, direction, title, description, category, price, and optional image. The listings endpoint rejects invalid values before inserting the record. A GET request retrieves active listings in reverse chronological order and maps database field names to the frontend model.

3.5.3 Updating and Delisting Listings

Users manage their own listings from the profile page. Delisting updates the delisted_at timestamp instead of permanently removing the record, which preserves a useful account history while excluding the post from active marketplace queries.

3.5.4 Security and Validation

The system combines client-side usability checks with server-side validation. Supabase row-level security restricts listing creation, update, and deletion to the record owner. Session tokens are kept on the server side, and the application does not expose database service-role credentials to the browser. Account deletion is implemented as an authenticated database function that can delete only the current user.

3.5.5 Implementation Limitations

- The current version does not provide online payment or transaction settlement.
- Communication is initiated through email rather than in-app messaging.
- There is no administrator dashboard or automated content moderation.
- The project depends on Supabase availability and correct deployment configuration.
- Image storage is limited to an encoded image in the listing record and has a size limit.

4. RESULTS AND DISCUSSION

4.1 Application Testing and Results

Testing was performed by checking normal user workflows, invalid input handling, authorization behavior, and persistence of records. The following test cases represent the main functional requirements.

Test ID	Test case	Expected result	Result
T01	Register with a valid college email	Account is created or confirmation is requested.	Pass
T02	Register with a non-college email	Request is rejected with a validation message.	Pass
T03	Sign in with valid credentials	Authenticated session is created and marketplace opens.	Pass
T04	Sign in with invalid credentials	User remains signed out and sees an error.	Pass
T05	View active marketplace listings	Listings are loaded from the backend.	Pass
T06	Search by keyword and select a category	Only matching listings are shown.	Pass
T07	Create an item listing	Validated listing is stored and displayed.	Pass
T08	Create a student-help request	Help listing is stored with request direction.	Pass
T09	Save and remove a favorite	Favorite state is persisted for the current user.	Pass
T10	Enquire about a listing	Gmail compose opens with the seller address.	Pass
T11	Delist one’s own post	Post disappears from active listings.	Pass
T12	Attempt invalid price or oversized image	Server rejects the request.	Pass
T13	Change password after current-password verification	Password is updated when validation succeeds.	Pass
T14	Delete account without confirmation	Deletion is refused.	Pass

4.1.1 Authentication Testing

Authentication tests confirmed that the application accepts only valid institutional email addresses, rejects malformed passwords, handles email confirmation, and does not expose the marketplace to unauthenticated users. Logout destroys the server session and returns the user to the login page.

4.1.2 Marketplace Testing

Marketplace tests confirmed that item and student-help listings are displayed in their appropriate tabs. Search and category filters update the visible cards, and the empty state is shown when no listing matches. Listing creation rejects values outside the required ranges.

4.1.3 Profile and Security Testing

Profile tests confirmed that a user can view and delist their own posts but cannot use the normal API flow to change another user’s records. Favorites are scoped to the current account. Password and account deletion workflows require current-password verification and explicit confirmation where appropriate.

4.2 Results and Discussion

The implemented system meets the core objectives of the project. It provides a coherent campus-specific browsing experience, supports two categories of community exchange, and persists records through a protected backend. The use of server-side PHP sessions and Supabase row-level security gives the prototype a stronger security model than a purely client-side application.

The main practical limitation is that the project currently connects buyers and sellers through email instead of an internal conversation system. The application also assumes that users and administrators will follow safe campus handover practices. These limitations are acceptable for the Web Programming Lab scope and provide clear directions for future enhancement.

5. CONCLUSION AND FUTURE WORK

Campify — Campus Marketplace was developed as a web application to make student-to-student exchange more organized, searchable, and campus-focused. The system integrates a responsive frontend with PHP APIs, Supabase authentication, and a PostgreSQL database. It supports institutional sign-in, listing creation, browsing, search, filtering, favorites, enquiry, delisting, password management, and secure account deletion.

The project demonstrates important web programming concepts including modular page design, asynchronous JavaScript, form validation, REST-style endpoints, server-side sessions, relational data modeling, and row-level security. Functional testing indicates that the core workflows behave as expected for valid and invalid inputs.

Future Enhancements

• Add an administrator dashboard for reporting, moderation, and removal of inappropriate listings.

• Provide in-app chat or masked contact communication between students.

• Add notifications for new listings, saved-listing changes, and enquiry responses.

• Introduce image storage with thumbnails and improved media management.

• Add ratings, seller verification, and reporting tools while protecting student privacy.

• Support payment integration only after appropriate institutional and security review.

• Develop a progressive web app or native mobile client.

• Add analytics and recommendations based on categories and campus demand.

REFERENCES

• MDN Web Docs, HTML, CSS, JavaScript, and Web APIs, https://developer.mozilla.org/

• PHP Documentation, PHP Manual, https://www.php.net/docs.php

• Supabase Documentation, Authentication and Database Guides, https://supabase.com/docs

• PostgreSQL Documentation, Row-Level Security, https://www.postgresql.org/docs/

• Project source files: index.html, login.html, profile.html, app.js, login.js, profile.js, api/, and schema.sql.

APPENDIX A — PROJECT FILE STRUCTURE

Path	Description
index.html	Main marketplace page
login.html	Sign-in and registration page
profile.html	Profile and account settings page
styles.css	Shared responsive styles
app.js	Marketplace interaction and listing rendering
login.js	Authentication form behavior
profile.js	Profile and account actions
api/	PHP endpoints for authentication, listings, favorites, profile, and delisting
schema.sql	Supabase database schema, policies, trigger, and account function
README.md	Setup and local execution instructions